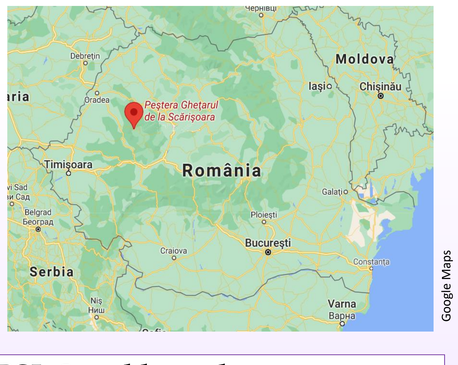
Cold active lipases from Scarisoara Ice Cave and their applicative potential

Ojovan BO[1], Gheorghita GR[2,1], Paun VI[1], Purcarea C[1]

1 - Institute of Biology, Department of Microbiology, Bucharest, Romania
2 - University of Bucharest, Faculty of Chemistry, Department of Organic Chemistry, Biochemistry and Catalysis, Bucharest, Romania





INTRODUCTION

Psychrophilic microorganisms can produce highly efficient cold-active enzymes (extremozymes) that are used as catalysts in various industries.
Our study focused on the lipolytic activity of seven bacterial strains isolated from ice deposits accumulated in Scarisoara Ice Cave, Romania. We cloned and characterized two lipases from the *Psychrobacter sp*, cave isolate:

- PSL-2 (483 aa, 53,6kDa)
- PSL-3 (315 aa, 34,6kDa)

Lipase activity of ice bacteria

Strain	Abbr.	Ice age (years BP)	Lipase activity (2- naphthyl-myristate)
Psychrobacter sp.	PSY	5.3k	4
Microbacterium sp.	MC	92	3
Pseudarthrobacter sp.	PA	10.1k	3
Arthrobacter sp.	AR	10.1k	1
Bacillus sp.	BA	10.1k	3
Brevundimonas sp.	BR	721	3
Candidimonas sp.	CA	456	1

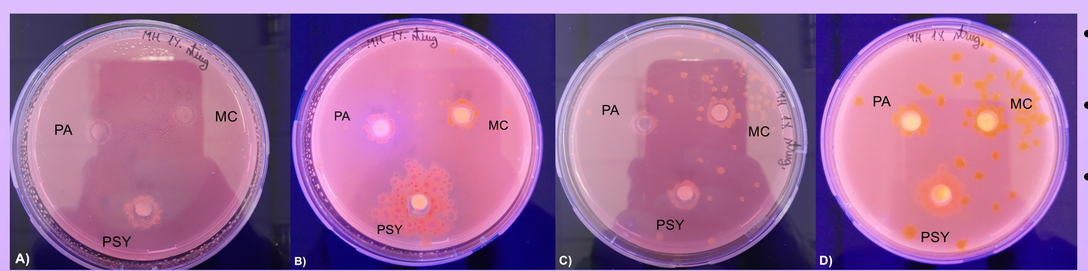
(Paun et al., 2021, *Scientific Reports* 11)

Identity and similarity scores between PSL-2 and PSL-3 and homologous enzymes

	Strains	PSL-2		PSL-3	
		Identity (%)	Similarity (%)	Identity (%)	Similarity (%)
Psychrophiles	*Psychrobacter sp G*	99	99	99	99
	Glaciibacter superstes	29	43	28	43
	Moritella sp PE36	28	43	28	42
Mesophiles	*Pseudomonas aeruginosa*	33	48	37	58
	Escherichia coli	28	43	26	43
Thermophiles	*Stenotrophomonas maltophilia*	30	41	55	57

Lipolytic activity on natural oils

Strain	Olive oil		Grape seed oil		Thistle oil	
	15°C	25°C	15°C	25°C	15°C	25°C
Psychrobacter sp.	+	+	+	+	+	
Microbacterium sp.	+	+	+	+	+	
Pseudarthrobacter sp.				+		+
Bacillus sp.						
Brevundimonas sp.	+	+	+	+	+	
Arthrobacter sp.	+					



Incubated screening plates: 15°C (A, C); 25°C (B, D)

Cold-active adaptation of PSL-2 and PSL-3

Homologs	Arg/ (Arg + Lys)	Homologs	Arg/ (Arg + Lys)
PSL-2	**0.39**	**PSL-3**	**0.26**
Glaciibacter superstes	0.82	*Glaciibacter superstes*	0.86
Moritella sp. PE36	0.47	*Moritella sp.* PE36	0.43
Escherichia coli	0.89	*Escherichia coli*	0.81
Pseudomonas aeruginosa	0.88	*Pseudomonas aeruginosa*	0.72
Stenotrophomonas maltophilia	1	*Stenotrophomonas maltophilia*	0.84

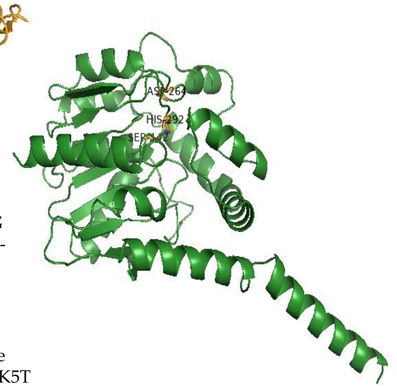
CONCLUSIONS

- Ice cave bacteria are able to hydrolyse olive oil, grape seed oil and thistle oil.
- PSL-2 and PSL-3 possess structural elements related to cold adaptation.
- Primary structure of PSL-2 and PSL-3 is homologous to cold-active lipase from Antarctic *Psychrobacter sp* G.
- PSL-2 and PSL-3 have similar folding with cold-active esterases from *Parvibaculum sp* and *Psychrobacter kryohalolentis* K5T respectively.

3D modelling of PSL-2 and PS-3 tertiary structure



PSL-2
Conserved motifs: HGGGF and GDSAG with Gly297
Catalytic triad: Ser299-Asp414-His444
Part of family IV or bacterial lipases
Template: esterase Est8 from *Parvibaculum* (Pereira et al., 2017, Appl Microbiol Biotechnol)



PSL-3
Conserved motifs: HGGGF and GNSMG
Catalytic triad: Ser142-Asp264-His292
Part of family V or bacterial lipases

Template: cold-active esterase from *Psychrobacter cryohalolentis* K5T

FINANCIAL SUPPORT: UEFISCDI PN-III-P2-2.1-PED-2019-2461